MARKET RISK · FRTB

Measuring market risk capital with the FRTB standardized approach

The theory behind the workbook: how sensitivities, defaults and residual risks become one capital number, block by block

Methodology guide to the K-FRTB-ASA workbook

Basel MAR21–MAR23 · status November 2026

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Sixteen parts, in calculation order

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Parts 4 to 15 follow the workbook’s calculation chain, from sensitivity rows on the input sheets to the total on the Results sheet.

ABOUT THIS GUIDE

What the guide covers, and in what order

01

The risk

Why positions marked to market every day need a capital charge of their own

02

The rulebook

Basel MAR chapters, the trading book boundary and how regions apply them

03

The engine

Sensitivities, risk weights, correlations and three scenarios for five risk classes

04

Non-linear risk

Vega for volatility, curvature for what delta and vega miss

05

Default and residual

Jump-to-default by obligor, and a flat add-on for risks the model cannot see

06

The limits

Where the formula misreads risk, and the readings the workbook had to decide

Losses that show up every day

Banking book

- Held to collect, usually at amortized cost
- Loss arrives through default or impairment
- Capital: credit risk framework (CRE)

Trading book

- Held for trading or hedging, fair valued daily
- Loss arrives through prices, before any default
- Capital: market risk framework (MAR)

Liquid

Positions can be sold or hedged, but not instantly and not at any size.

Linear and not

Bonds and swaps move with rates; options move with volatility and convexity too.

Correlated

Hedges offset in calm markets and can stop offsetting in stress.

PART 1 · WHY THE RULES CHANGED

The crisis exposed the old market risk rules

1996 · VaR

Market Risk Amendment: 10-day 99% value-at-risk, times a multiplier of at least 3, or a building-block standardized method

2009 · Basel 2.5

Stopgap after 2007–09: stressed VaR, the incremental risk charge and the comprehensive risk measure

2016–19 · FRTB

A full rewrite of both approaches: expected shortfall, liquidity horizons, a new boundary and a sensitivity-based standard

VaR said nothing about the size of tail losses, and the trading book attracted positions, such as credit structures, whose capital there was lower than in the banking book.

The standardized approach in this guide is the FRTB answer for every bank, not only for those without models.

BCBS market risk amendment (1996), Basel 2.5 (2009), FRTB: d352 (2016) and d457 (2019)

Five changes in one review

Area	Before FRTB	Under FRTB
Boundary	Intent-based, transfers easy	Presumptive lists, transfers restricted (MAR12)
Risk measure	10-day 99% VaR plus stressed VaR	97.5% expected shortfall, stress-calibrated
Liquidity	One 10-day horizon for everything	Horizons of 10 to 120 days by risk factor
Model approval	Bank-wide	Desk by desk, with P&L attribution and backtesting
Standard approach	Position-based building blocks	Sensitivities-based method, DRC and RRAO

The standardized approach no longer counts positions: it reads the same sensitivities a trading desk uses to hedge.

Three ways to compute market risk capital

MAR20–MAR23

Standardized approach

Sensitivities-based method, default risk capital and residual risk add-on. Required of every bank with a trading book above the thresholds.

MAR40

Simplified standardized

The Basel II building blocks, scaled up: 1.3 for rates, 3.5 for equity, 1.9 for commodities, 1.2 for FX. For small, simple books with supervisory approval.

MAR30–MAR33

Internal models

Expected shortfall per approved desk, a stressed charge for non-modelable risk factors and a model-based default charge.

A bank on internal models still computes the standardized approach for every desk: as a fallback, as the floor for the output floor and for disclosure.

One number, four regulatory uses

Use	What it does	Rule
Own funds	Market risk capital for banks without approved models	MAR20–MAR23
Fallback	Capital of a desk that fails the model tests	MAR32–MAR33
Credible benchmark	Computed for every desk, model-approved or not	MAR30, MAR33
Output floor	Total RWA at least 72.5% of standardized RWA	RBC20.11
Disclosure	Published capital by risk class and approach	DIS50

An error in the standardized engine therefore reaches even banks whose own funds come from internal models.

PART 1 · THE WORKBOOK

What the workbook calculates, and what it does not

Calculated

- Delta, vega and curvature for GIRR, CSR non-securitization, equity, commodity and FX
- Three correlation scenarios and the binding one
- Default risk capital, non-securitizations
- Residual risk add-on

Not calculated

- CSR for securitizations, non-CTP and CTP, and their DRC
- Sensitivities and CVRs: they are inputs, priced by the bank
- Which instruments carry an RRAO
- The simplified approach and internal models

Rows for an unsupported risk class are rejected and logged, so a missing component is never counted as zero.

The MAR chapters behind the workbook

Scope

MAR10–MAR12

Definitions, application, and the boundary between trading and banking book

The workbook

MAR20–MAR23

MAR21 sensitivities-based method, MAR22 default risk capital, MAR23 residual risk add-on

Models

MAR30–MAR33

Internal models: expected shortfall, desk tests, non-modelable risk factors, model DRC

MAR40

Simplified standardized approach

MAR50

CVA risk, a separate charge on derivative counterparties

MAR99

Guidance on the use of internal models

Basel Framework, MAR standard with FAQs (bis.org); MAR21 republished 23 March 2026 with FAQ changes

From first consultation to the capital rule

May 2012
First consultative paper, BCBS 219

Oct 2013
Second consultative paper, BCBS 265

Jan 2016
Minimum capital requirements for market risk, d352

Jan 2019
Revised standard, d457: recalibrated SA, simplified SA

Dec 2019
Consolidated into the Basel Framework as MAR

Mar 2020
Implementation deferred by one year, Covid-19

Jan 2023
Basel implementation date for the revised standard

2024–2027
Regional rules: EU CRR3, UK PRA, US agencies

BIS publications BCBS 219, 265, d352, d457; GHOS press release, 27 March 2020

The boundary decides what enters the calculation

Presumed trading book

Held for trading, listed equities, net short credit or equity positions, options, and instruments from underwriting or market making

Presumed banking book

Unlisted equities, real estate, retail and SME credit, funds without daily look-through

Reclassification

Only in extraordinary circumstances, approved by the supervisor; any capital reduction is disallowed

Internal risk transfers

Hedges from the banking book enter the trading book only through documented internal trades

One standard, several implementations

Region	Legal basis	Points to check
Basel	MAR, in force from 1 January 2023	The reference text for the workbook
EU	CRR as amended by CRR3, Regulation (EU) 2024/1623	FRTB own-funds start deferred by delegated acts; reporting under CRR2 since 2021
UK	PRA rules for Basel 3.1 market risk	Start dates moved more than once; check the current PRA policy
US	2023 agency proposal for the Basel III endgame	Final rule and timing set by the agencies

The workbook follows the Basel text. Its parameters are editable on the P_* sheets: check them against the rules that apply to you.

PART 3 · THE FORMULA

$Capital = SBM + DRC + RRAO$

SBM is the largest of three correlation scenarios, each the simple sum over risk classes of delta, vega and curvature. The three components are then added, with no diversification between them.

Basel Framework MAR20.4, MAR21.7

Three blocks feed one capital number

MAR21

Sensitivities-based method

Delta, vega and curvature for seven risk classes, aggregated with supervisory risk weights and correlations

MAR22

Default risk capital

Jump-to-default loss by obligor, netted by seniority, weighted by credit quality, with a hedge benefit ratio

MAR23

Residual risk add-on

Gross notional times 1.0% for exotic underlyings, 0.1% for other residual risks

3.16m

SBM of the sample portfolio, low scenario binding

0.64m

DRC, three buckets: corporates, sovereigns, local governments

0.08m

RRAO, three instruments

Sample portfolio total: 3,877,792.81 in the reporting currency.

From sensitivity rows to the capital number

1 Validate
Reject rows with unknown classes, buckets or sub-types, and log them

2 Net
Sum rows that share a risk factor

3 Weight
WS = risk weight × sensitivity

4 Bucket
Kb with correlations ρ, and Sb

5 Risk class
Across buckets with γ, alternative Sb if needed

6 Scenario
Repeat for low, medium, high; keep the largest

7 Total
Add DRC and RRAO

Inputs

In_Delta, In_Vega, In_Curvature, In_DRC, In_RRAO

Parameters

P_General, P_GIRR, P_CSR, P_EQ, P_COM, P_FX, P_DRC, P_RRAO

Outputs

Out_Buckets, Out_RiskClass, Out_DRC, Out_Summary, Results, Log

Workbook sheets; MAR21.4–MAR21.7

Seven risk classes, five in the workbook

Risk class	Bucket	Delta risk factor	Workbook
GIRR, general interest rate	Currency	Curve and tenor	Yes
CSR, non-securitization	18 sector × quality	Issuer, curve, tenor	Yes
CSR, securitization non-CTP	25 sector × quality	Tranche and tenor	No
CSR, correlation trading	16 sector × quality	Underlying name and tenor	No
Equity	13 size × region × sector	Spot price, repo rate	Yes
Commodity	11 commodity groups	Commodity, tenor, location	Yes
FX	Currency	Exchange rate to the reporting currency	Yes

Risk classes are added together: a long equity position and a short FX exposure never offset each other.

Offsets happen inside buckets first

Risk factor

The smallest unit: a curve tenor, a share price, a currency. Sensitivities to it net fully.

Bucket

Risk factors that move together, such as one currency. Offsets inside use correlations ρ.

Risk class

Buckets aggregate with correlations γ, lower than the ρ inside a bucket.

100%

offset between sensitivities to the same risk factor

7.5–99.9%

correlation ρ between risk factors of one bucket

0–75%

correlation γ between buckets of a risk class

Basel Framework MAR21.4; correlation ranges from MAR21.45–MAR21.89

Three correlation scenarios, the largest wins

$$high = min(1.25\,\rho, 1) \cdot medium = \rho \cdot low = max(2\rho - 1, 0.75\,\rho)$$

Applied to every ρ and γ; for curvature after squaring them

Prescribed correlation	Low	Medium	High
Two large-cap financial stocks, 25%	18.75%	25.00%	31.25%
Two FX buckets, 60%	45.00%	60.00%	75.00%
Brent and WTI, 95%	90.00%	95.00%	100.00%

Low correlation hurts hedged books, high correlation hurts directional books. The binding scenario is the largest total SBM, not chosen per risk class.

A delta is a scaled change in present value

Risk class	Shock to the risk factor	Delta sensitivity
GIRR	+1 bp to one tenor of one curve	ΔPV ÷ 0.0001
CSR	+1 bp to one tenor of the issuer’s spread curve	ΔPV ÷ 0.0001
Equity spot	+1% relative change in the share price	ΔPV ÷ 0.01
Equity repo	+1 bp to the repo rate	ΔPV ÷ 0.0001
Commodity	+1% relative change in the commodity price	ΔPV ÷ 0.01
FX	+1% relative change in the exchange rate	ΔPV ÷ 0.01

Dividing by the shock size turns the sensitivity into an exposure per unit of the factor: a 1% rate move on a GIRR delta of 5 million is a loss or gain of 50,000.

PART 4 · VEGA SENSITIVITIES

Vega is measured against implied volatility

$$s = vega \times implied\ volatility$$

vega = ∂V / ∂σ of the option, for each option maturity on the grid 0.5, 1, 3, 5, 10 years

Why times σ

The risk weight is a relative shock to volatility, so the input is the value change for a 100% relative move.

Two dimensions

Option maturity for every class; GIRR yield options also need the underlying's residual maturity.

Every option

Any instrument with optionality has vega, including callable bonds and prepayment options.

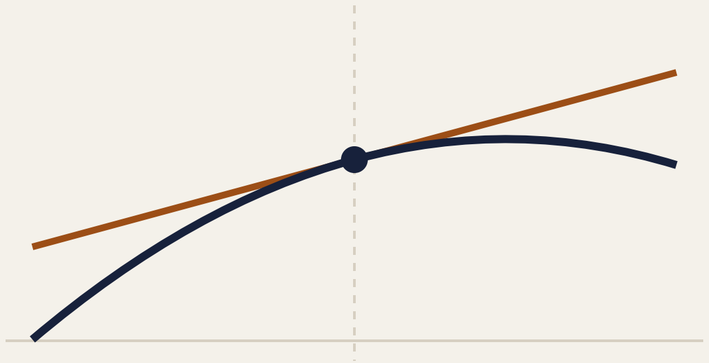
PART 4 · CURVATURE INPUTS

Curvature is the loss that delta does not explain

$$CVR^{+} = -\Sigma_i \left[V_i(x^{+}) - V_i(x) - RW \cdot s_i \right]$$

$$CVR^{-} = -\Sigma_i \left[V_i(x^{-}) - V_i(x) + RW \cdot s_i \right]$$

Full revaluation under an up and a down shock the size of the risk weight, net of the delta effect; positive means a loss



Long option

Beats its delta both ways: CVRs negative, no charge.

Short option

Loses more than its delta both ways: CVRs positive, a charge.

Basel Framework MAR21.5 · navy curve: short option value; copper line: delta tangent

Rows that share a risk factor are netted first

Row	Bucket	Curve	Tenor	Sensitivity	Net
D008	USD	SOFR	10y	2,000,000	
D009	USD	SOFR	10y	500,000	
	USD	SOFR	10y		2,500,000

36
delta rows in the sample portfolio

35
risk factors after netting: one pair shares a factor

100%
offset within a risk factor, before any correlation

A position and its exact hedge cancel here. Everything after this step recognizes only partial offsets.

PART 4 · INPUT SHEETS

One row per sensitivity, five columns that matter

Class	Bucket	Qualifier	Sub-type	Tenor
GIRR	Currency	Curve name	YIELD, INFLATION, XCCY	0.25 to 30 years
CSR_NS	1 to 18	Issuer or index	BOND, CDS	0.5, 1, 3, 5, 10
EQ	1 to 13	Issuer or index	SPOT, REPO	none
COM	1 to 11	Commodity	Delivery location	0 to 30 years
FX	Currency	Same as bucket	none	none

In_Delta

Plus the sensitivity in the reporting currency

In_Vega

Plus option maturity and, for GIRR yield, underlying maturity

In_Curvature

Plus CVR^{+}, CVR^{-} and, for FX, the reporting-currency flag

PART 4 · CONVENTIONS

Signs and units decide every offset

Reporting currency

All sensitivities, CVRs and notionals are entered in the reporting currency; the workbook converts nothing.

Sign

A positive delta gains when the risk factor rises. Opposite signs inside a bucket offset partly; across buckets they can add or offset.

FX direction

FX risk factors are the price of the foreign currency in the reporting currency. A long USD position has positive delta.

Curvature sign

The opposite convention: a positive CVR is a loss. A long option enters with negative CVRs.

One wrong sign turns a hedge into a doubled position. Sign conventions are the first thing to check in an input file.

PART 5 · WEIGHTED SENSITIVITY

A risk weight turns a sensitivity into a loss

$$WS_k = RW_k \times s_k$$

RW is the supervisory stress move of risk factor k, calibrated to its liquidity horizon

Sample risk factor	Net sensitivity	RW	WS
GIRR EUR, ESTR 5y	−12,000,000	1.1%	−132,000
CSR bucket 1, Italy bond 5y	3,000,000	0.5%	15,000
Equity bucket 8, Bank A spot	2,000,000	50%	1,000,000
Commodity bucket 2, Brent 0.5y	800,000	35%	280,000
FX, USD	3,000,000	15%	450,000

Basel Framework MAR21.4(3); workbook sample portfolio

PART 5 · INSIDE A BUCKET

Within a bucket, correlations allow partial offset

$$Kb = \sqrt{max(0, \Sigma_k WS_k^2 + \Sigma_k \Sigma_{l \neq k} \rho_{kl} WS_k WS_l)} \quad Sb = \Sigma_k WS_k$$

The floor at zero only matters for strongly hedged buckets; Sb keeps the sign for the next step

32,500

Sum of weighted sensitivities in CSR bucket 1: Italy bond, Spain bond, Italy CDS

26,738

Kb, medium scenario: name correlation 35%, bond-CDS basis 99.9%

32,500

What Kb would be with all correlations at 100%

Two different sovereigns are only 35% correlated, so a long in each diversifies; the same issuer’s bond and CDS are almost one risk.

Basel Framework MAR21.4(4), MAR21.54; workbook sample, CSR bucket 1

PART 5 · ACROSS BUCKETS

Buckets combine with weaker correlations

$$Delta = \sqrt{\left(\Sigma b \; Kb^2 + \Sigma b \; \Sigma c \neq b \; \gamma bc \; Sb \; Sc \right)}$$

Kb^2 keeps each bucket's own risk; the cross terms use the signed sums Sb, so longs and shorts in different buckets offset

Risk class	Cross-bucket correlation γ
GIRR	50% between currencies
CSR	Sector table × rating factor: 50% between investment-grade and high-yield named buckets
Equity	15% between names buckets, 75% between the two index buckets, 45% index with names
Commodity	20%
FX	60%

Basel Framework MAR21.4(5), MAR21.50, MAR21.57, MAR21.80, MAR21.85, MAR21.89

PART 5 · ALTERNATIVE SPECIFICATION

When hedges overshoot, Sb is capped at Kb

$$Sb = max(min(\Sigma_k WS_k , Kb), -Kb)$$

Used for every bucket of the risk class when the cross-bucket radicand is negative

CSR delta, high scenario	Kb	Σ WS	Sb used
Bucket 1, sovereigns IG	27,584	32,500	27,584
Bucket 2, local govt IG	30,475	−38,000	−30,475
Bucket 3, financials IG	5,000	5,000	5,000
Bucket 11, financials HY	12,000	12,000	12,000
Bucket 17, IG indices	12,000	−12,000	−12,000

Out_Buckets marks these rows ALT_SB, and Out_RiskClass shows Y under “Alternative Sb used”.

Basel Framework MAR21.4(5)(b); workbook sample, high scenario

PART 5 · THE OTHER-SECTOR BUCKETS

Other-sector buckets get no diversification

$$Kother = \Sigma_k \mid WS_k \mid Delta = \sqrt{(\ \Sigma\ Kb^2 + \Sigma\ \gamma\ Sb\ Sc\ )} + Kother$$

CSR bucket 16 and equity bucket 11: names that fit no sector, outside the square root

CSR bucket 16, sample

Long 200,000 and short 150,000 of two names at 12%: 24,000 + 18,000 = 42,000. The long and the short do not offset.

Effect on CSR delta

Medium scenario: 6,261 from the other buckets plus 42,000 from bucket 16 gives 48,261.

Putting a name in the other-sector bucket is the most expensive mapping: every position counts in full, whatever its sign.

Basel Framework MAR21.56, MAR21.79; workbook sample, CSR bucket 16

PART 5 · CHOOSING THE SCENARIO

The binding scenario depends on the book



Hedged books

Longs against shorts: lower correlation weakens the hedge, low binds.

Directional books

All long: higher correlation removes diversification, high binds.

One choice

The scenario is chosen on the total SBM, not per risk class.

Basel Framework MAR21.7; workbook sample, Out_Summary

One bucket per currency, three kinds of curve

YIELD

Risk-free curves

One curve per index, such as ESTR or EURIBOR 6M, at ten tenors: 0.25, 0.5, 1, 2, 3, 5, 10, 15, 20, 30 years

INFLATION

Inflation curve

One risk factor per inflation index, such as HICP, with no tenor dimension

XCCY

Cross-currency basis

One risk factor per currency against USD or EUR, with no tenor dimension

Sample, EUR bucket

ESTR at 1, 5 and 10 years; EURIBOR 6M at 5 and 10 years; HICP inflation

Sample, USD bucket

SOFR at 2 and 10 years; USD-EUR cross-currency basis

Basel Framework MAR21.8; workbook sample, In_Delta rows D001–D010

Short tenors move most in stress

Tenor	0.25	0.5	1	2	3	5	10	15	20	30
Risk weight	1.7%	1.7%	1.6%	1.3%	1.2%	1.1%	1.1%	1.1%	1.1%	1.1%

1.6%

inflation and cross-currency basis risk factors

÷ √2

bank's discretion for specified currencies: EUR, USD, GBP, AUD, JPY, SEK, CAD and the reporting currency

1.20%

the 1.7% weight after the √2 discretion

The workbook applies the √2 division to the inflation and basis weights too, the literal reading of "the above risk weights" in MAR21.44.

Basel Framework MAR21.42–MAR21.44; Control sheet, GIRR_SPECIFIED_CCY_DISCRETION

Nearby tenors move together

$$\rho_{kl} = max(exp(-3\% \cdot |T_k - T_l| / min(T_k, T_l)), 40\%)$$

Same curve; × 99.9% across curves of one currency; 40% inflation against yield; 0% with the cross-currency basis

Tenor pair, same curve	ρ, medium
5y and 10y	97.0%
1y and 5y	88.7%
1y and 10y	76.3%
0.25y and 30y	40.0% (floor)

50%

γ between two currencies

99.9%

factor between two curves of one currency, also between two inflation curves

PART 6 · WORKED EXAMPLE

The EUR bucket: six risk factors, one charge

Risk factor	Sensitivity	RW	WS
ESTR 1y	5,000,000	1.6%	80,000
ESTR 5y	−12,000,000	1.1%	−132,000
ESTR 10y	8,000,000	1.1%	88,000
EURIBOR 6M 5y	3,000,000	1.1%	33,000
EURIBOR 6M 10y	−1,500,000	1.1%	−16,500
HICP inflation	400,000	1.6%	6,400

58,900
Sb, the signed sum

50,673
Kb, medium scenario

356,000
sum of absolute WS: the charge with no offsets at all

Workbook sample, In_Delta D001–D006; Out_Buckets, GIRR delta EUR

Two currencies, partly offsetting

$$\sqrt{(\ 50{,}673^2 + 34{,}346^2 + 2 \times 50\% \times 58{,}900 \times (-8{,}500)\ )} = 56{,}980$$

EUR bucket: Kb 50,673, Sb 58,900 · USD bucket: Kb 34,346, Sb −8,500 · γ = 50%, medium scenario

60,424	**56,980**	53,315
GIRR delta, low scenario	GIRR delta, medium scenario	GIRR delta, high scenario

USD: SOFR 2y −52,000, SOFR 10y +27,500 after netting two rows, USD-EUR basis +16,000. The net short USD bucket offsets part of the net long EUR bucket.

Eighteen buckets by sector and credit quality

Sector	IG	HY & NR
Sovereigns, central banks	1 · 0.5%	9 · 2%
Local govt, public	2 · 1%	10 · 4%
Financials	3 · 5%	11 · 12%
Materials, energy	4 · 3%	12 · 7%
Consumer, transport	5 · 3%	13 · 8.5%
Technology, telecom	6 · 2%	14 · 5.5%
Health, utilities	7 · 1.5%	15 · 5%

Special bucket	RW
8 · Covered bonds, IG	2.5%
16 · Other sector	12%
17 · IG indices	1.5%
18 · HY indices	5%

Financials carry the highest weights: 5% investment grade, 12% high yield.

Name, tenor and basis multiply

$$\rho_{kl} = \rho name \times \rho tenor \times \rho basis$$

Each factor is 1 when the two risk factors share that dimension

35%

Different names; 80% between two indices in buckets 17 and 18

65%

Different tenors of the same name

99.9%

Bond curve against CDS curve of the same name

Across buckets: γ = sector correlation from Table 5 × rating factor, 50% between an investment-grade and a high-yield named bucket, otherwise 100%.

PART 7 · WORKED EXAMPLE

A bond hedged with CDS on the same bank

Bucket 3, Bank A, 3y	Sensitivity	RW	WS
Bond spread	1,500,000	5%	75,000
CDS spread	−1,400,000	5%	−70,000

$$Kb = \sqrt{(\ 75{,}000^2 + 70{,}000^2 - 2 \times 99.9\% \times 75{,}000 \times 70{,}000\ )} = 5{,}958$$

Same name, same tenor: only the 0.1% bond-CDS basis stops a full offset

145,000
charge with no offset

5,958
Kb, medium scenario

5,000
Sb, the net long

Workbook sample, In_Delta D016–D017; Out_Buckets, CSR delta bucket 3

PART 7 · CSR DELTA IN THE SAMPLE

Six buckets, and a scenario surprise

Bucket	Kb, medium	Sb	Note
1 · Sovereigns IG	26,738	32,500	long
2 · Local govt IG	29,796	−38,000	short
3 · Financials IG	5,958	5,000	basis
11 · Financials HY	12,000	12,000	long
16 · Other sector	42,000	6,000	outside √
17 · IG indices	12,000	−12,000	short

62,844
low scenario

48,261
medium scenario: the lowest

60,572
high scenario, alternative Sb

Workbook sample, Out_Buckets and Out_RiskClass, CSR delta

Thirteen buckets: size, region, sector

Large cap, by sector	Emerging	Advanced
Consumer, transport, health, utilities	1 · 55%	5 · 30%
Telecom, industrials	2 · 60%	6 · 35%
Materials, energy, mining	3 · 45%	7 · 40%
Financials, real estate, tech	4 · 55%	8 · 50%

Other buckets	Spot RW
9 · Small cap, emerging	70%
10 · Small cap, advanced	50%
11 · Other sector	70%
12 · Large-cap advanced indices	15%
13 · Other indices	25%

Large cap means a market capitalization of at least USD 2 billion; advanced economies are listed by the Basel Committee.

Basel Framework MAR21.72–MAR21.77, Tables 9 and 10

Two risk factors per name, one correlation table

SPOT

Share price

Delta for a 1% relative move. Risk weight 15% to 70% by bucket.

REPO

Repo rate

Delta for a 1 bp move in the equity repo rate. Risk weight = spot weight ÷ 100.

Correlation between two names in a bucket	ρ
Large cap, emerging (1–4)	15%
Large cap, advanced (5–8)	25%
Small cap, emerging (9) · advanced (10)	7.5% · 12.5%
Indices (12, 13)	80%
Spot against repo, same name	99.9%

PART 8 · WORKED EXAMPLE

Long one bank, short another: little offset

Bucket 8, financials, advanced	Sensitivity	RW	WS
Bank A spot	2,000,000	50%	1,000,000
Bank B spot	−1,200,000	50%	−600,000
Bank A repo	50,000	0.5%	250

$$Kb \approx \sqrt{(\ 1{,}000{,}000^2 + 600{,}000^2 - 2 \times 25\% \times 1{,}000{,}000 \times 600{,}000\ )} = 1{,}029{,}769$$

Exact figure includes the repo terms; medium scenario

400,250
Sb, the net long

1,029,769
Kb, medium

1,065,572
Kb, low: 18.75% correlation

Workbook sample, In_Delta D022–D024; Out_Buckets, EQ delta bucket 8

An index short hedges single names partly

Bucket	Kb	Sb	γ with 8
5 · Consumer, advanced	270,000	270,000	15%
8 · Financials, advanced	1,029,769	400,250	
12 · Large-cap indices	225,000	−225,000	45%
11 · Other sector	70,000	70,000	0%, outside √

1,156,963
equity delta, low

1,109,540
equity delta, medium: 1,039,540 + 70,000

1,059,848
equity delta, high

Basel Framework MAR21.80; workbook sample, Out_RiskClass, EQ delta

Eleven commodity groups, each with its own weight

Bucket	RW	ρ
1 · Energy, solid	30%	55%
2 · Energy, liquid	35%	95%
3 · Electricity, carbon	60%	40%
4 · Freight	80%	80%
5 · Non-precious metals	40%	60%
6 · Gaseous combustibles	45%	65%

Bucket	RW	ρ
7 · Precious metals	20%	55%
8 · Grains, oilseed	35%	45%
9 · Livestock, dairy	25%	15%
10 · Softs, other agri	35%	40%
11 · Other commodity	50%	15%

ρ is the correlation between two different commodities of the bucket: Brent and WTI share 95%, two livestock contracts only 15%.

Commodity, tenor and delivery location multiply

$$\rho_{kl} = \rho commodity \times \rho tenor \times \rho location$$

Each factor is 1 when the two risk factors share that dimension

Commodity

15–95%

bucket-specific, Table 11

Tenor

99%

two maturities, 0 to 30 years

Location

99.9%

two delivery locations

Across buckets γ = 20%, and 0% with bucket 11. Commodities are the least correlated asset class across groups.

A calendar spread in oil, plus gold and power

Risk factor	Sensitivity	RW	WS
2 · Brent, Rotterdam, 0.5y	800,000	35%	280,000
2 · Brent, Rotterdam, 1y	−500,000	35%	−175,000
2 · WTI, Cushing, 1y	300,000	35%	105,000
3 · Power DE, EEX, 0.25y	200,000	60%	120,000
7 · Gold, London, spot	1,000,000	20%	200,000

208,323
Kb of bucket 2, medium; Sb 210,000

366,440
commodity delta, medium, γ 20%

379,605
commodity delta, high: this class prefers high

Workbook sample, In_Delta D028–D032; Out_Buckets and Out_RiskClass, COM delta

One risk factor per currency, one weight for all

15%
risk weight for every currency against the reporting currency

10.6%
15% ÷ √2 for specified pairs and their first-order crosses, at the bank's discretion

60%
γ between two currency buckets

Bucket = currency

Each bucket holds a single risk factor, so Kb = |WS| and all offsetting happens across buckets.

Specified pairs

USD against EUR, JPY, GBP, AUD, CAD, CHF, MXN, CNY, NZD, RUB, HKD, SGD, TRY, KRW, SEK, ZAR, INR, NOK, BRL; also EUR/JPY, EUR/GBP, EUR/CHF, JPY/AUD.

MAR21.14, MAR21.86–MAR21.89, fn 23; Control: FX_SPECIFIED_PAIR_DISCRETION

PART 10 · WORKED EXAMPLE

Four currencies, computed by hand

Currency	Delta	WS = Sb	Kb² (thousands²)
USD	3,000,000	450,000	202,500
GBP	−1,500,000	−225,000	50,625
JPY	800,000	120,000	14,400
CHF	−400,000	−60,000	3,600

In thousands: Σ Kb^2 = 271,125 and Σ of the six products Sb·Sc = −94,950, so FX delta = √(271,125 − 189,900 γ).

430,894
low, γ = 45%

396,466
medium, γ = 60%

358,748
high, γ = 75%

Workbook sample, In_Delta D033–D036; Out_RiskClass, FX delta

PART 10 · PRACTICAL POINTS

What counts as one currency

Currency variants

Onshore and offshore, deliverable and non-deliverable forms of a currency are one risk factor: enter KRW for every won sensitivity.

Base currency

With supervisory approval a bank may measure FX against a base currency instead of the reporting currency.

Every FX exposure

FX delta includes foreign-currency bonds, equities and commodities in the trading book, not only FX derivatives.

The workbook rejects an FX row with a sub-type, and treats every currency code as its own bucket.

Basel Framework MAR21.14 and FAQ; MAR21.24

Liquidity horizons set the vega weights

$$RW = min(55\% \times \sqrt{(LH \div 10)}, 100\%)$$

LH is the regulatory liquidity horizon of the risk class, in days

Risk class	LH, days	Vega RW
GIRR	60	100%
CSR non-securitization	120	100%
Equity, large cap and indices (1–8, 12, 13)	20	77.78%
Equity, small cap and other (9–11)	60	100%
Commodity	120	100%
FX	40	100%

Basel Framework MAR21.92, Table 13, and FAQ on equity indices

PART 11 · VEGA CORRELATIONS

Option maturities decay slowly

$$\rho_{kl} = min(\, \rho option \times \rho underlying\, ,\, 1\,)\quad \rho option = exp(\, -1\% \cdot |T_k - T_l| \,/\, min(T_k, T_l)\,)$$

ρ underlying: the delta correlation of the underlying risk factors; for CSR and equity the name only, for commodity the commodity only

99.0%

6-month against 1-year options on the same underlying

96.1%

1-year against 5-year options

γ

across buckets: the same γ as delta

Swaptions live on a two-dimensional surface

EUR vega row	Option	Underlying	Vega × σ	WS, RW 100%
V001 swaption	1y	5y	400,000	400,000
V002 swaption	5y	10y	−250,000	−250,000

$$\rho = exp(-1\% \times 4 \div 1) \times exp(-1\% \times 5 \div 5) = 0.9608 \times 0.9900 = 95.1\%$$

Kb = √(400,000² + 250,000² − 2 × 95.1% × 400,000 × 250,000) = 179,594, medium

YIELD

Option maturity × underlying residual maturity

INFLATION

Option maturity only, one per inflation curve

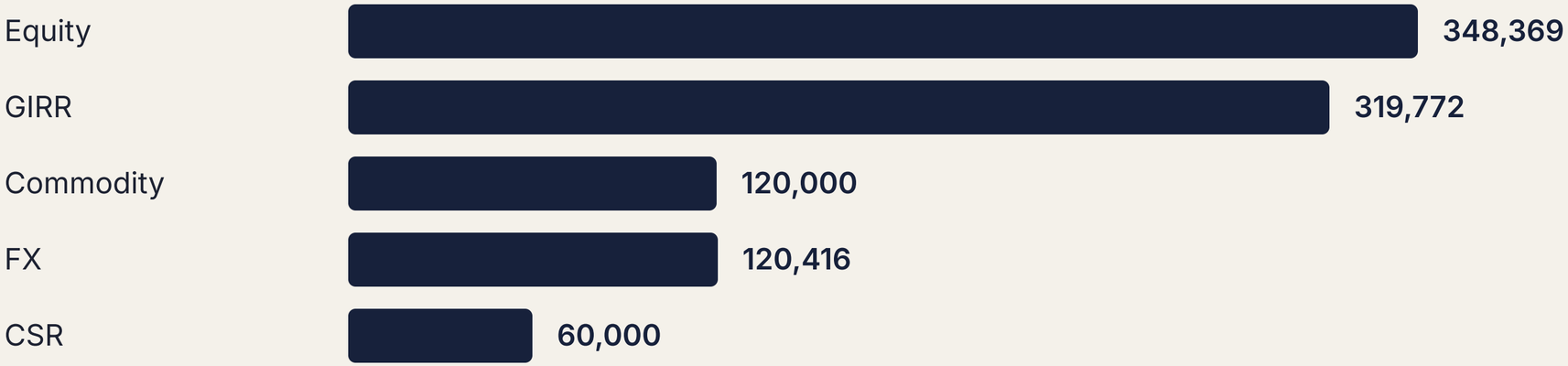
XCCY

Option maturity only

Basel Framework MAR21.8 and FAQ4, MAR21.93; workbook sample, In_Vega V001–V002

PART 11 · VEGA IN THE SAMPLE

Vega is half as large as delta in the sample



968,557
total vega, medium scenario

1,005,692
total vega, low scenario, the binding one

Workbook sample, Out_RiskClass, vega; bars at the medium scenario

Coarser risk factors, a full revaluation

Risk class	Curvature risk factor	What moves together
GIRR	One per currency	All yield curves, all tenors; no inflation or basis
CSR	One per issuer	Bond and CDS curves, all tenors
Equity	One per issuer or index	Spot price only
Commodity	One per commodity	All tenors and delivery locations
FX	One per currency	Exchange rate

Delta and vega are local measures. Curvature captures large moves of the whole curve or price, where options pay off non-linearly.

The pricing system applies the shocks

Risk class	Up and down shock	Delta deducted
FX, equity	Relative move equal to the delta risk weight	Its delta to the risk factor
GIRR, CSR	Parallel shift of every tenor by the bucket's highest delta weight; GIRR uses 1.7%	Sum of deltas to all tenors of the curves
Commodity	Every tenor price × (1 ± RW), RW of the bucket; gold at 20%: × 1.2 and × 0.8	Sum of deltas to all tenors and locations

CSR floor

If the down shock makes a spread negative, it may be floored at zero, capping the weight in CVR^{-} only.

√2 and the shock

MAR21 does not say whether the √2 discretions reduce the curvature shock. The workbook takes no position.

Up and down are aggregated separately

$$Kb^{+} = \sqrt{max(0, \Sigma\, max(CVR_k^{+}, 0)^2 + \Sigma\,\Sigma\, \rho_{kl}^2\, CVR_k^{+}\, CVR_l^{+}\, \psi)} \cdot Kb = max(Kb^{+}, Kb^{-})$$

$$Curvature = \sqrt{max(0, \Sigma\, Kb^2 + \Sigma\,\Sigma\, \gamma bc^2\, Sb\, Sc\, \psi(Sb, Sc))}$$

ψ = 0 when both arguments are negative, otherwise 1 · Sb = Σ CVR of the scenario the bucket selected

Squared correlations

ρ² and γ² replace ρ and γ, then the three scenarios apply.

Gains count once

A gain offsets a loss only through a cross term; two gains never combine.

Worse side wins

Each bucket keeps the larger of its up and down charges.

FX curvature can be divided by 1.5

FX_CURVATURE_DIVISOR

Standard discretion

Divide the CVRs of options that do not reference the reporting currency. Flag the others TRUE in “References reporting ccy”.

FX_CURVATURE_DIVISOR_ALL

Needs approval

Divide every FX CVR, if the supervisor approves and curvature covers shocks to the reporting currency too.

Sample row	CVR^{+}	CVR^{-}	Result
USD, flag TRUE	−40,000	−35,000	Kb 0, Sb −35,000, down side
GBP	15,000	12,000	Kb 15,000, Sb 15,000, up side

FX curvature of the sample is zero: the USD gain offsets the GBP loss at $\gamma^2 = 36\%$, and the radicand is negative.

PART 12 · WORKED EXAMPLE

Equity curvature, step by step

Risk factor	CVR^+	CVR^-	ρ^2 or γ^2
Bucket 8 · Bank A	−80,000	40,000	ρ^2 = 6.25%
Bucket 8 · Bank B	20,000	−60,000	
Bucket 12 · EURO STOXX 50	30,000	−10,000	γ^2 = 20.25%

Bucket 8 up

√(20,000² − 2 × 6.25% × 80,000 × 20,000) = 14,142

Bucket 8 down

√(40,000² − 2 × 6.25% × 40,000 × 60,000) = 36,056, Sb −20,000

Bucket 12 up

Kb 30,000, Sb 30,000

$$\sqrt{(\,36{,}056^2 + 30{,}000^2 + 2 \times 20.25\% \times (-20{,}000) \times 30{,}000\,)} = 44{,}238$$

Workbook sample, In_Curvature C004–C006; Out_RiskClass, EQ curvature, medium scenario

A default is a jump, not a small move

What

Loss if the obligor defaults today, on bonds, CDS, equities and their derivatives in the trading book

How

Jump-to-default by position, scaled for maturity, netted by obligor and seniority, weighted by credit quality

Where

Three buckets for non-securitizations: corporates, sovereigns, local governments and municipalities

Credit spread sensitivities measure the cost of spreads widening. DRC measures the loss when they stop mattering because the issuer has defaulted.

Loss given default, plus the mark-to-market

$$JTD_{long} = max(\, LGD \times notional + P\&L,\, 0\,) \quad JTD_{short} = min(\, LGD \times notional + P\&L,\, 0\,)$$

P&L = market value − notional, signed; notional negative for shorts; then × min(max(M, 0.25), 1)

Seniority	LGD
Equity, non-senior debt	100%
Senior debt	75%
Covered bonds	25%

Italy, sample row J001

Senior bond, notional 10,000,000, market value 9,800,000, 5 years: 75% × 10,000,000 − 200,000 = 7,300,000, scale 1.

Maturity scaling: below one year the JTD is cut in proportion, with a floor of three months. Cash equity is either three months or at least one year.

PART 13 · NETTING BY OBLIGOR

A short offsets a long only if it ranks the same or lower

Bank A, sample	Seniority	Maturity	Scaled JTD
J003 senior bond	Senior	3y	3,050,000
J004 shares	Equity	1y	2,000,000
J005 short senior	Senior	0.5y	−1,135,000
Net long			3,915,000

Seniority rule

The senior short offsets the senior bond only. It cannot hedge the shares, which rank below it; a short in shares could hedge any long.

One bucket, worst rating

An obligor split across buckets is rejected; differing ratings resolve to the worst, with a warning.

Basel Framework MAR22.19–MAR22.24; workbook sample, In_DRC J003–J005; workbook methodology

Shorts hedge longs, scaled by the hedge benefit ratio

$$HBR = \Sigma\ net\ long \div (\Sigma\ net\ long + \Sigma\ |net\ short|)$$

$$DRCb = max(\Sigma\ RW \times net\ long - HBR \times \Sigma\ RW \times |net\ short| , 0)$$

Total DRC = simple sum of the buckets; no offset between corporates, sovereigns and local governments

Quality	AAA	AA	A	BBB	BB	B	CCC	Unrated
Risk weight	0.5%	2%	3%	6%	15%	30%	50%	15%

A book that is half short gets an HBR of one half: shorts can never offset more than the share of longs in the bucket. Defaulted names carry 100%.

Basel Framework MAR22.22–MAR22.26, Table 2

Eight obligors after netting

Obligor	Bucket	Quality	RW	Net long	Net short
Bank A	CORP	A	3%	3,915,000	
Bank B	CORP	BBB	6%		1,450,000
Consumer X	CORP	BB	15%	175,000	
Misc Co	CORP	Unrated	15%	355,000	
Bank Z	CORP	CCC	50%	100,000	
Italy	SOV	BBB	6%	7,300,000	
Germany	SOV	AAA	0.5%		3,850,000
Region X	LGM	AA	2%	1,510,000	

Consumer X: 0.1-year maturity, scaled by the 0.25 floor to 175,000. Bank Z: covered bond, LGD 25%.

PART 13 · DRC BY BUCKET

A cheap short hedges an expensive long poorly

Bucket	HBR	Σ RW × long	Σ RW × short	DRC
CORP	75.8%	246,950	87,000	180,993
SOV	65.5%	438,000	19,250	425,397
LGM	100%	30,200	0	30,200

636,589

DRC for non-securitizations, the sum of the three buckets

Sovereigns

Short 3.85 million of Germany against long 7.3 million of Italy removes only 12,603: a AAA short weighs 0.5%, a BBB long 6%.

A flat charge for what sensitivities miss

1.0% of gross notional

Exotic underlyings

Longevity, weather, natural disasters, future realized volatility: risks with no delta, vega or curvature factor in MAR21.

0.1% of gross notional

Other residual risks

Gap risk, correlation risk, behavioral risk such as prepayment: payoffs the sensitivities cannot replicate.

Exempt

Listed and cleared instruments, back-to-back trades

Gross

No netting and no diversification

Classified upstream

The workbook takes the category as an input

PART 14 · WORKED EXAMPLE

Three instruments, one line of arithmetic

Sample instrument	Category	Gross notional	RW	RRAO
Weather derivative	EXOTIC	5,000,000	1.0%	50,000
Correlation basket option	OTHER	20,000,000	0.1%	20,000
Behavioral prepayment option	OTHER	7,500,000	0.1%	7,500
Total		32,500,000		77,500

77,500

RRAO of the sample, 2% of the total capital

65%

of the add-on comes from 15% of the notional: the exotic weather trade

Workbook sample, In_RRAO R001–R003; P_RRAO

PART 15 · THE SAMPLE PORTFOLIO

A hedged multi-asset book in 68 rows

36
delta rows: rates, credit, equity, commodity, FX

10
vega rows: swaptions, equity, oil and FX options

9
curvature rows, one per risk factor

13
default and residual-risk rows

Hedged by design

Long against short in most buckets: a calendar spread, a basis trade, single names against an index.

Built to test edges

It hits the alternative Sb, an other-sector bucket, a zero FX curvature and the three-month DRC floor.

Reproducible

It ships in every build: press Run FRTB-SA, then Run regression test.

Workbook sample, In_* sheets; 52 regression items in Test_Expected

Equity dominates, curvature is small

Low scenario	Delta	Vega	Curvature	Total
GIRR	60,424	323,277	33,354	417,055
CSR	62,844	60,000	5,000	127,844
Equity	1,156,963	367,879	45,747	1,570,589
Commodity	352,785	120,000	10,000	482,785
FX	430,894	134,536	0	565,431
SBM	2,063,910	1,005,692	94,101	3,163,703

GIRR is the one class where vega exceeds delta: the EUR and USD swaptions carry more risk than the rate deltas, which hedge each other along the curve.

Workbook sample, Out_RiskClass, low scenario; totals may differ by 1 from rounding

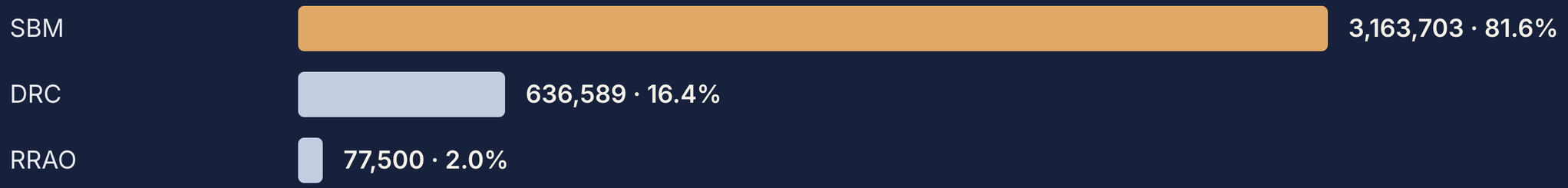
Not every class prefers the binding scenario

Delta + vega + curvature	Low	Medium	High
GIRR	417,055	410,664	404,003
CSR	127,844	113,261	125,572
Equity	1,570,589	1,502,148	1,430,225
Commodity	482,785	496,440	509,605
FX	565,431	516,882	463,151
SBM	3,163,703	3,039,394	2,932,556

Commodity is highest under high correlation, CSR under low, yet both are charged at low: the scenario is chosen on the total.

PART 15 · FROM CAPITAL TO RWA

3.88 million of capital, 48.5 million of RWA



3,877,793
capital requirement, in the reporting currency

48,472,410
risk-weighted assets: capital × 12.5

Workbook sample, Results and Out_Summary; RWA = capital × 12.5 (RBC20)

PART 16 · KNOWN LIMITATIONS

Where the formula misreads risk

No cross-class offset

An equity hedge of a credit position, or a rates hedge of FX, earns nothing.

Fixed correlations

Three scenarios widen the range but stay linear; tail dependence is not modeled.

Cliff effects

A name moving bucket, or a rating crossing into high yield, changes weights abruptly.

Coarse curvature

Two shocks per risk factor, no cross-gamma between factors.

Single-name hedges

25% or less between two names: relative-value trades look almost unhedged.

Data burden

Sensitivities for every position and the shock contract for every option.

Common critiques of standardized market risk approaches; see the further reading

Six interpretations, recorded and implemented

Question	Decision of October 27, 2026
GIRR √2 scope	Also divides the 1.6% inflation and basis weights (MAR21.44)
Covered bonds AA-	2.5% kept; an override changes the whole bucket 8
FX curvature divisor	Per-row flag; the divide-all variant labeled as needing approval
DRC obligor data	Split buckets rejected; worst rating applies; cash equity 0.25 or ≥ 1 year
Scenario ties	Order low, medium, high, with a note on Results
Curvature weights	CVRs stay inputs; the shock contract is documented

None of these changes the sample results: the discretions are off and the sample already complies.

An independent reference, two hosts, one Excel record

1
BIS check
Every parameter compared with the BIS text; 59 FAQs reviewed

2
Hand cases
Small examples worked from the MAR formulas

3
Reference
Python, written separately from the VBA

4
Reconcile
VBA run in LibreOffice equals the reference on every output

5
Excel
21 of 21 checks passed in desktop Excel, November 2026

52
regression items embedded in the workbook

21 / 21
unattended Excel checks at the recorded commit

0
regulatory validation: results are not certified

Workbook validation.md and evidence records; Excel check of 12 November 2026

PART 16 · KEY TAKEAWAYS

Seven things to remember

1 · Capital = SBM + DRC + RRAO, with no diversification between them.

2 · Inputs are sensitivities the desk already computes, in the reporting currency.

3 · Offsets are full within a risk factor, partial in a bucket, weaker across buckets, nil across classes.

4 · Three correlation scenarios; the largest total binds, chosen once for all classes.

5 · Curvature charges what delta misses, from full revaluation under the risk-weight shock.

6 · In DRC a short hedges only within its seniority and is scaled by the hedge benefit ratio.

7 · Every number of the sample can be reproduced: 3,877,792.81 in total.

Four results to reproduce by hand

FX delta, medium

396,466

√(271,125 − 189,900 × 0.60) thousand, from four currencies at 15% (slide 49)

Equity bucket 8, medium

1,029,769

Bank A long, Bank B short, 25% name correlation (slide 43)

DRC corporates

180,993

246,950 − 75.8% × 87,000, hedge benefit ratio applied (slide 65)

RRAO

77,500

5,000,000 × 1.0% + 27,500,000 × 0.1% (slide 67)

REFERENCES

Primary sources

Source	Content
Basel Framework, MAR10–MAR12	Definitions, application, trading book boundary
Basel Framework, MAR20–MAR23	Standardized approach: SBM, DRC, RRAO; MAR21 republished March 2026
Basel Framework, MAR30–MAR33, MAR40	Internal models; simplified standardized approach
MAR standard with FAQs, bis.org	The FAQs on MAR21–MAR23 used in this guide
BCBS d457, January 2019	Minimum capital requirements for market risk, revised
BCBS d352, January 2016	The original FRTB standard
Regulation (EU) 2024/1623	CRR3, including the EU market risk rules

Check the current versions on bis.org and in the official journal of your jurisdiction

REFERENCES

The workbook and further reading

Workbook documentation

- methodology.md: formulas, parameters and decisions
- validation.md: what is checked, and what is not
- workbook-guide.md: building and using the workbook
- Excel evidence records, October and November 2026

Background

- BCBS 219 (2012) and BCBS 265 (2013): the consultations and their rationale
- BCBS explanatory note on the revised market risk framework (2019)
- BCBS, Basel 2.5 revisions to the market risk framework (2009)

The workbook’s figures are not validated for regulatory use. Verify every parameter against the rules that apply to you.